

DEPARTMENT OF COMPUTERE ENGINEERING
SUBJECT: Machine Learning

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Experiment no. 7	
AIM :	Exp 7 : K means Clustering
Problem Statement:	The objective of this experiment is to apply K-Means clustering on a customer dataset to identify distinct groups based on features such as income, spending score, age, and gender. Additionally, the goal is to determine the optimal number of clusters using the Elbow Method and Silhouette Score.
Theory	K-Means clustering is an unsupervised machine learning algorithm used to partition a dataset into K distinct clusters based on feature similarity. The algorithm works by initializing K centroids, assigning each data point to the nearest centroid, and then recalculating the centroids as the mean of assigned points. This iterative process continues until convergence, where cluster assignments no longer change significantly. One of the key challenges in K-Means is selecting the optimal number of clusters. The Elbow Method helps address this by plotting the Within-Cluster Sum of Squares (WCSS) against different values of K. The point where the decrease in WCSS slows down significantly, forming an “elbow,” indicates a suitable number of clusters. Another method is the Silhouette Score, which measures how similar a data point is to its own cluster compared to other clusters. A higher silhouette score indicates better-defined and well-separated clusters. Feature scaling is an important preprocessing step in K-Means because the algorithm relies on distance calculations. Without scaling, features with larger ranges can dominate the clustering process. Overall, K-Means is widely used in customer segmentation, pattern recognition, and data analysis due to its simplicity and efficiency.
CODE:	<pre>import numpy as np import pandas as pd import matplotlib.pyplot as plt from sklearn.cluster import KMeans</pre>

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```
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler

!pip install xlrd openpyxl

# df = pd.read_excel('/content/sample_data/Mall_Customers.xls',
engine='xlrd')
df = pd.read_csv('/content/sample_data/Mall_Customers.xls')

df.head()

df = df.drop(['CustomerID'], axis=1)

df['Gender'] = df['Gender'].map({'Male': 0, 'Female': 1})

scaler = StandardScaler()
X = scaler.fit_transform(df)

wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('WCSS')
plt.show()

sil_scores = []

for k in range(2, 30):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X)
    score = silhouette_score(X, labels)
```

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```
sil_scores.append(score)

plt.plot(range(2, 30), sil_scores, marker='o')
plt.title('Silhouette Scores')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Score')
plt.show()

best_k = range(2, 30)[np.argmax(sil_scores)]
print("Optimal K (Silhouette):", best_k)

kmeans = KMeans(n_clusters=best_k, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X)

df['Cluster'] = clusters
df.head()

plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'], c=clusters)

plt.xlabel('Annual Income')
plt.ylabel('Spending Score')
plt.title('Customer Segments')

plt.show()

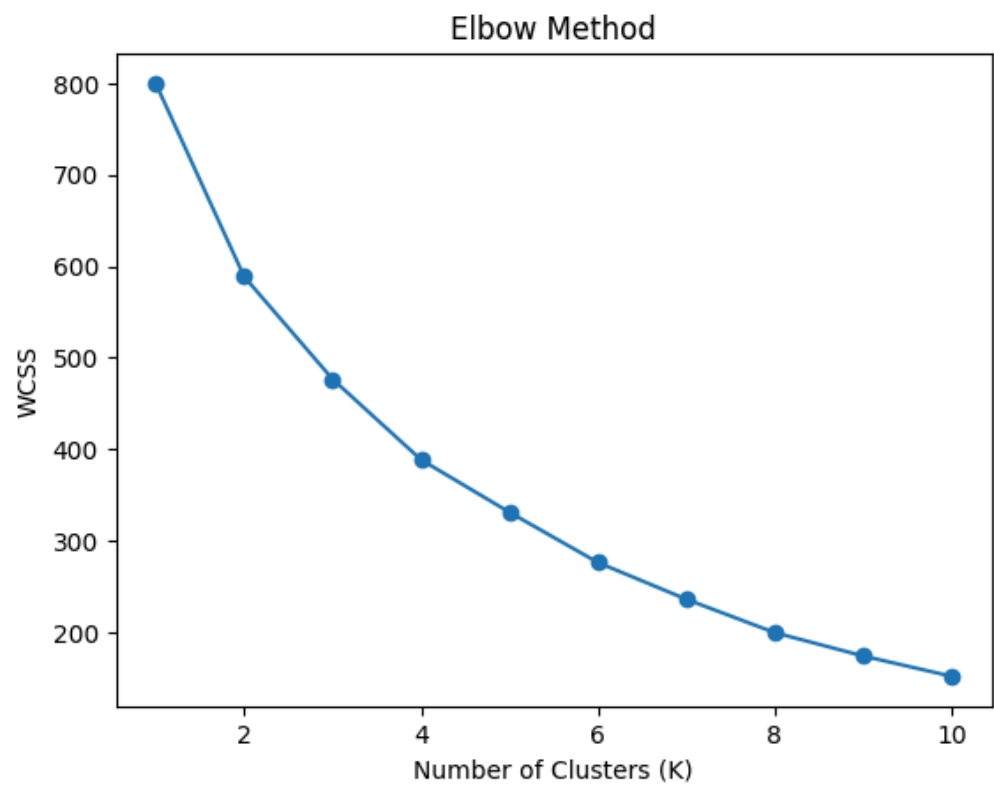
centers = scaler.inverse_transform(kmeans.cluster_centers_)
print(pd.DataFrame(centers, columns=df.columns[:-1]))

plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'], c=clusters,
label='Data Points')
plt.scatter(centers[:,2],centers[:,3], marker='X', s=200, label="Centroids",
color='red')
plt.title('Customer Segments with Centroids')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.show()
```

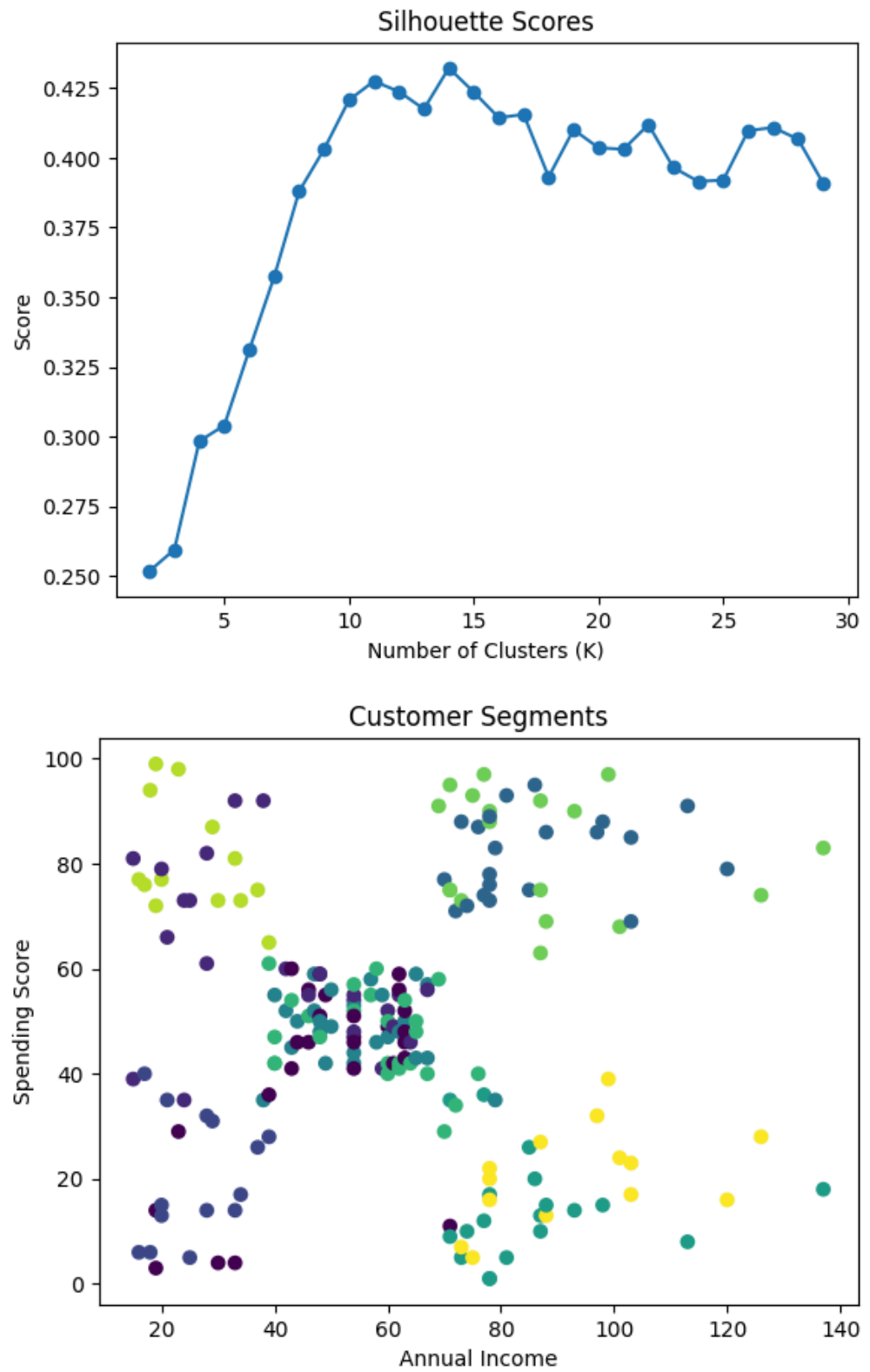
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OUTPUT:

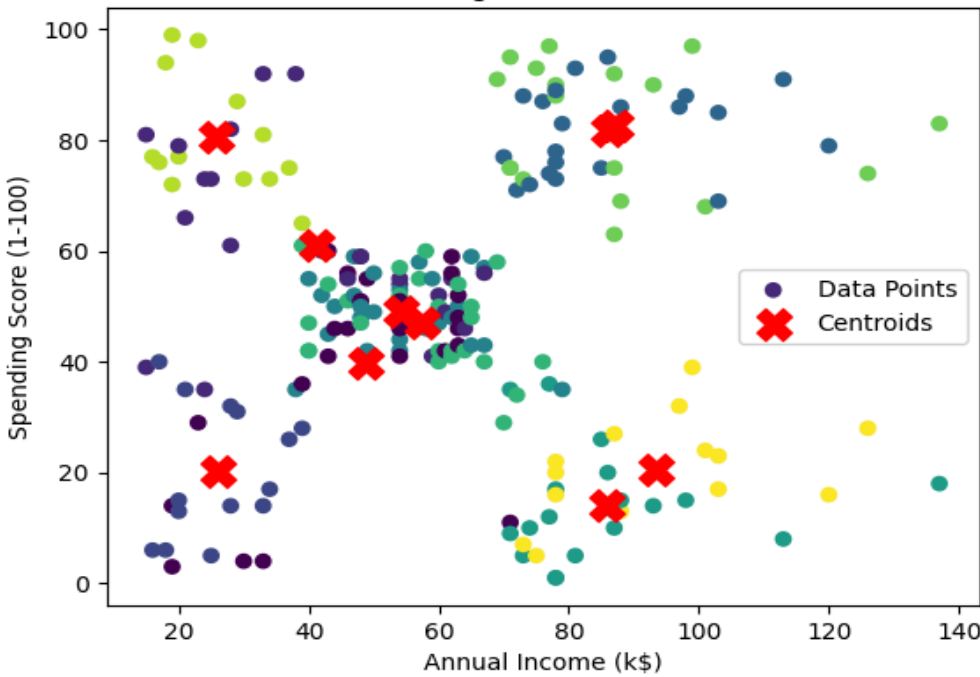
	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40



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	Customer Segments with Centroids  The scatter plot displays customer segments based on Annual Income (k\$) on the x-axis (ranging from 20 to 140) and Spending Score (1-100) on the y-axis (ranging from 0 to 100). Data points are represented by colored dots, and centroids are marked with red 'X's. The plot shows five distinct clusters of customers, each with its own centroid. The clusters are roughly located at: (20, 80), (40, 60), (60, 50), (80, 80), and (100, 20).
CONCLUSION:	In this experiment, K-Means clustering was successfully applied to segment customers into meaningful groups based on their characteristics. By using both the Elbow Method and Silhouette Score, an optimal number of clusters was determined, ensuring effective grouping and improved interpretability of results. The preprocessing step of feature scaling played a crucial role in achieving accurate clustering outcomes. The results demonstrate how unsupervised learning techniques can uncover hidden patterns in data without prior labeling. Customer segmentation obtained from clustering can help businesses understand different customer behaviors, enabling targeted marketing strategies and better decision-making. Although K-Means is simple and computationally efficient, it has limitations such as sensitivity to initial centroid selection and the need to predefine the number of clusters. Overall, this experiment highlights the practical importance of clustering techniques in real-world applications and reinforces the role of data preprocessing and evaluation metrics in achieving reliable machine learning results.
IPYNB FILE:	 EXP7.ipynb

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